MEDTHREAD

Clinical Intelligence Workspace

Implementation, Scientific Evaluation & Submission Guide

Project detail	Verified description
Application	React 18 / Vite / TypeScript frontend; FastAPI / Python backend
NLP	Shared deterministic Python pipeline; legacy Streamlit entry point retained
Report date	27 October 2026
Validation	141 backend tests passed; frontend production build succeeded

IMPORTANT: MedThread is an educational prototype using synthetic demo notes. It is not a diagnostic tool, provides no treatment recommendations, has no clinical validation or protected-health-information guarantees, and must not be used to make clinical decisions.

Compiled from the project source, documentation, tests, and observed API results.

Contents

1. Executive summary and architecture
2. Product pages and user workflows
3. Shared NLP pipeline and evidence semantics
4. Evaluation methodology and recomputed results
5. API and data handling
6. Validation and reproducibility
7. Repository file map
8. Scientific limitations and next steps
9. Academic references and scope

Scores in this report are synthetic development checks, not clinical or held-out performance.

1. Executive summary and architecture

MedThread is a local workspace for exploring clinical-note text and reviewing rule-generated longitudinal evidence. React/Vite/TypeScript calls FastAPI; FastAPI invokes reusable NLP modules under src/medthread/. The legacy Streamlit application remains an optional entry point.

- **Import:** preview/validate CSV or JSON, manually enter a note, filter active notes, restore demo data.
- **Explore:** inspect original note text, entity highlights, patient timelines, concept counts and co-occurrences.
- **Investigate:** inspect longitudinal flags with traceable evidence and session-only review choices.
- **Evaluate:** compute context metrics over source-aligned explicit labels; show coverage limits and unscorable predictions.
- **Scope:** synthetic-only corpus, no fitted ML model, no patient-group holdout, no clinical validation.

Current benchmark recomputation: 60 gold rows; 59 matched; 0 eligible missed extractions; 1 unscorable sentence alignment; 0 excluded gold rows. Eight unmatched predictions are preserved as unscorable, not false positives.

Main components

Layer	Responsibilities	Representative paths
Frontend	Navigation, screens, filters, evidence views, user state	frontend/src/pages/*.tsx; frontend/src/api/client.ts
FastAPI	REST routes, validation, active in-memory notes, exports	backend/main.py; backend/models.py; backend/state.py
NLP	Extraction, context, temporal, longitudinal, mining, evaluation	src/medthread/*.py
Data	Synthetic notes, partial context labels, benchmark provenance	data/*.csv; data/*.json
Verification	API and NLP regression tests; project docs	tests/*.py; docs/*.md

Data flow

```
CSV / JSON / manual note
-> backend field/date/duplicate validation
-> preprocessing and sentence splitting
-> lexicon entity extraction (regex; optional spaCy PhraseMatcher)
-> context classification
-> best-effort temporal extraction and normalization
-> patient + concept grouped longitudinal rules
-> API views, review evidence, mining, exports and evaluation
```

2. Product pages and user workflows

Page / route	Implemented behavior
Clinical Home / /	Corpus summary and recent review signals.
Clinical Notes / /import	CSV/JSON upload, preview counts/errors, manual entry, loaded-note list/filter, synthetic demo restore.
Patient Journey / /journey	Actual patient selector, chronological visits, mention status/context, source note evidence, original text highlighting.

Page / route	Implemented behavior
Note Explorer / /notes	Searchable notes, original text and inline entity highlights, selected mention evidence and temporal context.
Consistency / /consistency	Expandable evidence comparison, notes/dates/sentences/contexts, rule trace, temporal distinctions, session review actions.
Medical Concept Explorer (Text Intelligence) / /intelligence	Patient/date/context filters, concept counts/context distributions, note-level co-occurrence and source evidence.
Evaluation / /evaluation	Benchmark provenance, computed metrics, supports/matrix, errors, exclusions/unscorable predictions, unavailable-task explanations.

Import rules

- CSV columns: patient_id, note_id, visit_date, text. JSON is an array of records with those fields.
- Manual fields: patient ID, note ID, visit date, note text. Backend validates records again.
- Preview displays valid/invalid counts and row errors before import. File imports replace the active set; manual entry appends.
- Accepted import is processed by the same pipeline. Active data is in memory only and is lost when the backend stops.
- File size limit is 10 MB. Use only synthetic or properly de-identified content in this prototype.

Patient Journey

- Patient selection is based on the loaded dataset; records are grouped by patient.
- Expanding a visit shows original note text and extracted highlights.
- Each mention shows note ID/date, source sentence, context, rule explanation and available time interpretation.
- Newly documented means first seen in the available record, not biological onset.
- A missing mention means no lexicon match in that note, never proof of absence or resolution.

Consistency review

- Evidence comparison links exact sentences to source notes and dates and shows extraction contexts/reasons.
- Rules distinguish temporal changes from potential contradiction; flags are review signals, not conclusions.
- Actions: Confirmed inconsistency, Not a contradiction, Needs further review.
- Decisions are session-level prototype annotations; they do not update clinical records. CSV export is available.

Concept exploration

- Frequencies are extracted mention counts, not patient prevalence.
- Co-occurrence means two concepts were extracted in one note; it does not imply causation, diagnosis or clinical relationship.
- Selections expose note IDs and exact evidence sentences; empty/small results do not use fabricated values.

3. Shared NLP pipeline and evidence semantics

Entity extraction

Configured aliases map to canonical entities. The system uses word-boundary regex matching, or optional spaCy PhraseMatcher, with longest-first overlap suppression. Mention rows retain sentence text, offsets, extraction method and context evidence for UI traceability. Finite lexicon coverage can miss unseen aliases.

Context labels

Label	Meaning and guardrail
affirmed	Text states a current/ongoing mention; partial improvement is not resolution.
negated	Explicit denial in the applicable clause; cue scope is heuristic.
uncertain	Possible, suspected or hedged mention; not upgraded to affirmation.
resolved	Explicitly ended; “no longer” can resolve. Silence does not.
unclear	Ambiguous or insufficient text. Missing mention is not a negative.

Context rules isolate clauses and apply cue priorities for explicit resolution, negation, uncertainty and default affirmation. These are textual labels, not verified medical truth.

Temporal and longitudinal rules

The temporal module recognizes limited absolute/relative date forms, weekdays, month/day expressions and event-anchored dates. Normalization is best-effort relative to visit date; unresolved expressions remain unnormalized. Longitudinal comparisons group by patient ID and concept. Different normalized event dates can indicate temporal change rather than contradiction. Missing mentions are not treated as absence.

Safe rendering and storage

- React renders note content as text, not injected markup.
- Imported text is not emitted into application logs by the documented import flow.
- Active notes/results are backend process memory; reviewer choices are browser-session scoped.
- This prototype has no production authentication, durable storage, PHI security guarantee or clinical audit controls.

4. Evaluation methodology and recomputed results

Benchmark and alignment

Benchmark medthread-synthetic-context-v1 v0.1.0 contains 15 synthetic notes across 5 patient groups and 60 project-authored context rows. Labels are partial, not exhaustive and not independently reviewed. There are no entity offsets, exhaustive negative/no-entity labels, temporal gold labels or longitudinal case labels.

Alignment key: note ID + canonical entity + punctuation-insensitive source sentence tokens. The evaluation API disables unique-entity fallback so a prediction from a different sentence is not silently assigned to a gold row. Only aligned matches contribute to context metrics; eligible missed extractions are reported separately.

Per-class metrics and sample support

Gold context	Precision	Recall	F1	Support
Affirmed	1.000	1.000	1.000	25
Negated	1.000	1.000	1.000	23
Uncertain	1.000	1.000	1.000	4
Resolved	1.000	1.000	1.000	7
Unclear	0.000	0.000	0.000	0

Metric / coverage	Observed value
Aligned context examples	59
Accuracy	1.000
Macro precision / recall / F1	0.800 / 0.800 / 0.800
Weighted precision / recall / F1	1.000 / 1.000 / 1.000
Annotated-example entity recall	1.000; 59 matched / 59 eligible
Extraction precision / F1	Unavailable; annotations are partial
Gold coverage	59 matched; 0 eligible misses; 1 sentence mismatch; 0 excluded rows
Unmatched predictions	8 unscorable: 7 no note/entity gold row; 1 sentence mismatch
Scored context errors	0 returned
Temporal / longitudinal metrics	Unavailable; no gold labels for these tasks

Confusion matrix

Rows = gold labels; columns = predicted labels. Order: affirmed, negated, uncertain, resolved, unclear.

Gold / predicted	Affirmed	Negated	Uncertain	Resolved	Unclear
Affirmed	25	0	0	0	0
Negated	0	23	0	0	0
Uncertain	0	0	4	0	0
Resolved	0	0	0	7	0
Unclear	0	0	0	0	0

Why macro F1 differs: The fixed label set has five classes. “Unclear” has zero support; zero-division behavior assigns it score 0, which contributes one fifth of the unweighted macro mean. Weighted F1 weights each class by support, so the zero-support class has no weight. The four supported classes are perfect on these 59 examples, giving weighted F1 1.000. This is a label support/policy effect, not evidence of representative real-world performance.

Extraction metrics: The reported 1.000 is only an annotated-example matched/eligible ratio and is not corpus-wide recall. Valid extraction precision/recall/F1 require an explicitly scoped exhaustive annotation of all target entity occurrences, exact character offsets, repeated mentions, no-entity regions, documented coverage, independent review/adjudication, and a stated span matching rule. The current annotations cannot verify false positives.

Unmatched predictions: Eight outputs are retained in the audit: seven lack a note/entity gold row; one has a sentence alignment mismatch. They are reported as unscorable, not false positives, because the annotation set is not exhaustive.

Temporal and longitudinal gaps: Temporal evaluation needs reviewed event-expression/date/status labels, including current, historical, onset, resolved and ambiguous cases. Longitudinal evaluation needs patient-grouped note pairs, comparable time context, source IDs, rationale and labels for contradiction, temporal change, resolution, missing documentation, uncertainty and not comparable, with adjudication.

Holdout: There is no model fit, no patient-group train/test split and no untouched final holdout. The complete small corpus supports development checks only.

5. API and data handling

Endpoint group	Representative routes
Health/demo/stats	GET /api/health, /api/state, /api/stats; POST /api/demo
Notes/import	GET /api/patients, /api/notes, /api/notes/{id}; POST /api/notes/preview and /api/notes/import
Mentions/journey	GET /api/mentions, /api/timeline/{patient_id}, /api/lexicon
Consistency/review	GET /api/comparison, /api/review; PUT /api/review/{key}
Concepts	GET /api/concepts, /api/cooccurrence
Evaluation	GET /api/evaluation/demo
Exports	GET /api/export/mentions.csv, /api/export/review.csv

- Backend revalidates note fields and dates. CSV/JSON upload preview is capped at 10 MB.
- API errors and loading/empty states are surfaced to the frontend.
- CSV export escapes formula-like values.
- No durable database is implemented; process restart clears active imported data.

6. Validation and reproducibility

Check	Observed result
Backend/NLP/API suite	python -m pytest tests -q: 141 passed (2026-10-27).
Frontend	Set-Location frontend; npm run build: succeeded; TypeScript build and Vite production bundling.
Evaluation API	Recomputed supports, fixed confusion matrix, 8 unscorable predictions, no gold exclusions.
API import scenario	Previewed and imported two notes; checked note detail, timeline, review flag and concepts; restored demo and rechecked 59 samples.
Browser workflow	Import/manual/loaded pages, Note Explorer, Patient Journey, expanded review evidence, Concept Explorer and Evaluation inspected.

Observed environment

Tool	Version
Python / FastAPI	3.15.0 / 0.162.0
pandas / Pydantic / scikit-learn	3.0.6 / 2.14.1 / 1.9.1
pytest	9.1.1
Node.js / npm	v24.22.0 / 11.19.1
React / Vite / TypeScript	18.3.1 / 5.4.21 / 5.9.3

Run on Windows PowerShell

```
py -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r backend\requirements.txt
python -m pip install pytest
python -m uvicorn backend.main:app --host 127.0.0.1 --port 8001
```

```
Set-Location frontend
npm ci
npm run dev
```

Open http://localhost:5173. FastAPI documentation: http://127.0.0.1:8001/docs.

```
python -m pytest tests -q
Set-Location frontend
npm run build
```

7. Repository file map

Path	Purpose
backend/main.py	REST endpoints, import lifecycle, pipeline calls, exports and evaluation response.
backend/models.py; backend/state.py	Pydantic response contracts; in-memory corpus and reviewer states.
src/medthread/config.py	Lexicon and context rule configuration.
src/medthread/validation.py; preprocessing.py	Input checks, CSV/JSON parsing, normalization and sentence segmentation.
src/medthread/entity_extraction.py	Alias matching, canonical entity assignment and overlap handling.
src/medthread/context_detection.py	Affirmation, negation, uncertainty, resolution and unclear rules.
src/medthread/temporal.py	Time phrase extraction and best-effort normalization relative to visit date.
src/medthread/longitudinal_analysis.py	Patient-scoped comparison rules and review flags.
src/medthread/text_mining.py; evaluation.py	Concept aggregation; alignment, metrics and annotation coverage audit.
frontend/src/pages/*.tsx	Import, journey, note inspector, investigation, concept explorer, evaluation.
frontend/src/api/client.ts	Typed API client, session header and CSV downloads.
data/	Synthetic notes, partial gold labels, metadata and annotation schema.
tests/	Import, API, extraction, context, temporal, longitudinal, mining and evaluation tests.
docs/; README.md	Architecture, project/annotation/evaluation report, reproducibility, demo, viva and related work.

Path	Purpose
app.py; pages/	Retained legacy Streamlit application.

8. Scientific limitations and prioritized next steps

Priority	What remains before stronger claims
1. Reporting	Keep scores labeled as synthetic development checks, never clinical or held-out performance.
2. Extraction evaluation	Build an exhaustive offset-based entity corpus with explicit scope, no-entity regions, adjudication and span matching.
3. Temporal/longitudinal evaluation	Add reviewed temporal events and paired case labels with time comparability and distinct change/contradiction/resolution categories.
4. Holdout	If estimating generalization, reserve patient/case groups before tuning and keep final holdout untouched. None exists today.
5. Annotation quality	Add independent annotators, adjudication, agreement reporting and complete benchmark provenance.
6. Clinical deployment	Before real clinical use, require security/privacy threat modeling, authentication, authorization, secure storage, audit, and regulatory review.
7. Literature claims	Related Work is an orientation list, not a systematic search; verify citations and make no novelty claim.

Additional technical limits: finite lexicon, heuristic context/clause scope, best-effort date parsing, in-memory state, no authentication, and no independently adjudicated benchmark. Co-occurrence does not establish a clinical relationship.

9. Academic reference scope

The repository's Related Work document is an orientation bibliography, not a systematic review. It reports no novelty or research-gap claim. MedThread does not reproduce these systems, use their datasets, or compare performance against them.

Reference	Relevance
Chapman et al. (2001), NegEx, Journal of Biomedical Informatics 34(5), 301-310. DOI 10.1006/jbin.2001.1029	Negation rules.
Harkema et al. (2009), ConText, Journal of Biomedical Informatics 42(5), 839-851. DOI 10.1016/j.jbi.2009.05.002	Negation, experiencer and temporal status.
Uzuner et al. (2011), 2010 i2b2/VA challenge, JAMIA 18(5), 552-556. DOI 10.1136/amiajnl-2011-000203	Established concept/assertion/relation tasks.
Bethard et al. (2015), SemEval-2015 Task 6: Clinical TempEval, pp. 806-814. DOI 10.18653/v1/S15-2136	Clinical temporal evaluation.
Bethard et al. (2017), SemEval-2017 Task 12: Clinical TempEval, pp. 565-572. DOI 10.18653/v1/S17-2093	Clinical temporal evaluation.

Full references and scope caveats: docs/RELATED_WORK.md. Do not claim clinical validity, diagnosis, causality, prevalence or novelty.